

E-Commerce Management System

Module: Review and Rating Management Module — End User & Technical Documentation

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GitHub Repo: ansarizeenat/ecommerce-category-management	Live Server (Render): https://ecommerce-category-management.onrender.com
Tech Stack: Java 25, Spring Boot 4.1, JPA, MySQL 8.0, Thymeleaf	Date: September 2026

1. Introduction

The **Review and Rating Management Module** enables customers to leave feedback on products they have purchased. This feature helps prospective customers make well-informed purchasing decisions based on genuine customer experiences and allows businesses to gather actionable insights about product quality and customer satisfaction.

To preserve review integrity and protect the platform against spam, abuse, or inappropriate content, all newly submitted customer reviews default to an **unapproved (pending moderation)** state. Administrators review each submission and selectively approve or reject them before they become visible to the public.

2. Database Design & Specification

Table Name: **reviews** — strictly matches the Itvedant assignment specification.

Table 1: reviews

Column Name	Data Type	Key / Constraint	Description
review_id	INT	PK, Auto-Increment	Unique identifier for each review
product_id	INT	FK (products.id)	Foreign key referencing the products table
customer_id	INT	FK (users.user_id)	Foreign key referencing the users / customers table
rating	INT	NOT NULL (1 to 5)	Rating given by the customer (1 to 5 stars)
review_text	VARCHAR(1000)	NULLABLE	Textual feedback provided by the customer
status	BOOLEAN	DEFAULT FALSE	True for approved reviews, False for unapproved/pending
verified_purchase	BOOLEAN	DEFAULT FALSE	True if customer purchased this item in confirmed order
created_at	DATETIME	AUTO-GENERATED	Timestamp when the review was created
updated_at	DATETIME	AUTO-UPDATED	Timestamp when the review was last updated

3. Key Functionalities

3.1 Add a Review & Rating

- Customers can add a review and rating for products they have purchased.
- The system allows selecting a score between 1 and 5 stars with an interactive star selector.
- Provides textual feedback of up to 1000 characters with live character counter.
- **Purchase Verification:** Confirms that the customer has placed an order containing the item.
- **Default Moderation:** Reviews automatically default to unapproved (status = False) until approved by Admin.

3.2 Review Moderation

- Administrators can moderate reviews, approving or rejecting them based on quality and adherence to guidelines.
- Moderation dashboard allows filtering by status (All, Pending Moderation, Approved) and by product.
- One-click Approval marks status = True, instantly publishing it to the public storefront.
- Reject / Unapprove toggles status back to False, hiding it from public display.

3.3 View Product Reviews & Rating Breakdown

- Customers can view reviews and ratings for each product on its product page / details view.
- Computes the average rating (e.g. 4.8 / 5.0) rounded to 1 decimal place.
- Provides a visual star distribution breakdown showing progress bars and percentages for 1 to 5 stars.
- Displays reviewer name, verified purchase badge, star rating, creation date, and review text.

3.4 Delete / Update Reviews

- Admins can delete inappropriate, profane, or abusive reviews at any time.
- Customers can update or delete their own reviews.
- When an approved review is updated by a customer, its status automatically reverts to Pending Moderation.

4. End User Guide

Customer Workflow:

1. Navigate to the **Review & Rating** tab.
2. Select your Customer account and the Product you wish to review.
3. Click on the 1-5 gold stars to set your rating score.
4. Write your experience in the feedback box and click **Submit Review & Rating**.
5. The review is submitted and queued for Admin moderation.

Admin Moderation Workflow:

1. Open the **Review Moderation & Management Dashboard** at the bottom of the Reviews tab.
2. Click **Pending Moderation** filter to view newly submitted reviews.
3. Click **Approve** to publish the review, or **Reject** if it violates guidelines.
4. Inappropriate submissions can be permanently removed with the **Delete** button.

5. REST API Endpoints Reference

Method	Endpoint	Description	Access
GET	/api/reviews	List all reviews with optional status/productId filters	Public / Admin
GET	/api/reviews/stats	Platform statistics: totals, approved, pending, avg rating	Public / Admin
GET	/api/reviews/{id}	Fetch review by primary key ID	Public / Admin
GET	/api/reviews/product/{id}	Get public approved reviews for a product	Public
GET	/api/reviews/product/{id}/summary	Average rating score and 1-5 star breakdown bars	Public
POST	/api/reviews	Submit review (default status: unapproved / pending)	Customer
PUT	/api/reviews/{id}/moderate	Moderate review: approve (true) or reject (false)	Admin
PUT	/api/reviews/{id}	Update review (resets status to unapproved)	Customer / Admin
DELETE	/api/reviews/{id}	Delete review (admin or owner authorization)	Customer / Admin

6. Automated Testing & Verification

A suite of 9 JUnit 5 Spring Boot integration tests was created in ReviewServiceTest.java:

- Validated review creation, 1 to 5 star rating boundaries, and default unapproved status.
- Verified purchase enforcement preventing unpurchased item reviews without permission.
- Verified admin moderation (Approve / Reject workflow) and public visibility toggling.
- Verified average rating calculation and percentage distribution calculation.
- Verified customer update resetting status to pending, and admin delete privileges.

Suite Result: 49 / 49 tests passing across all system modules (0 failures, 0 errors).